WASP-52b

R-1652

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_200929` · created 2026-08-02T08:18:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459121.7558 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.063
Transit depth (Rp/Rs)^2	4.65 +/- 2.73 [%]
Orbital Inclination (inc)	85.43 +/- 2.73
Airmass coefficient 1 (a1)	0.118 +/- 0.051
Airmass coefficient 2 (a2)	1.77 +/- 0.87
Scatter in the residuals of the lightcurve fit is	41.17 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.042 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.063 vs 0.1646 (deviation 0.65σ)
Duration fit vs published	0.042 d vs 0.0758 d (deviation 2.26σ)
Mid-time O–C	-13.8 min
Depth SNR	2.8
Residual scatter	41.17 %

Light curve

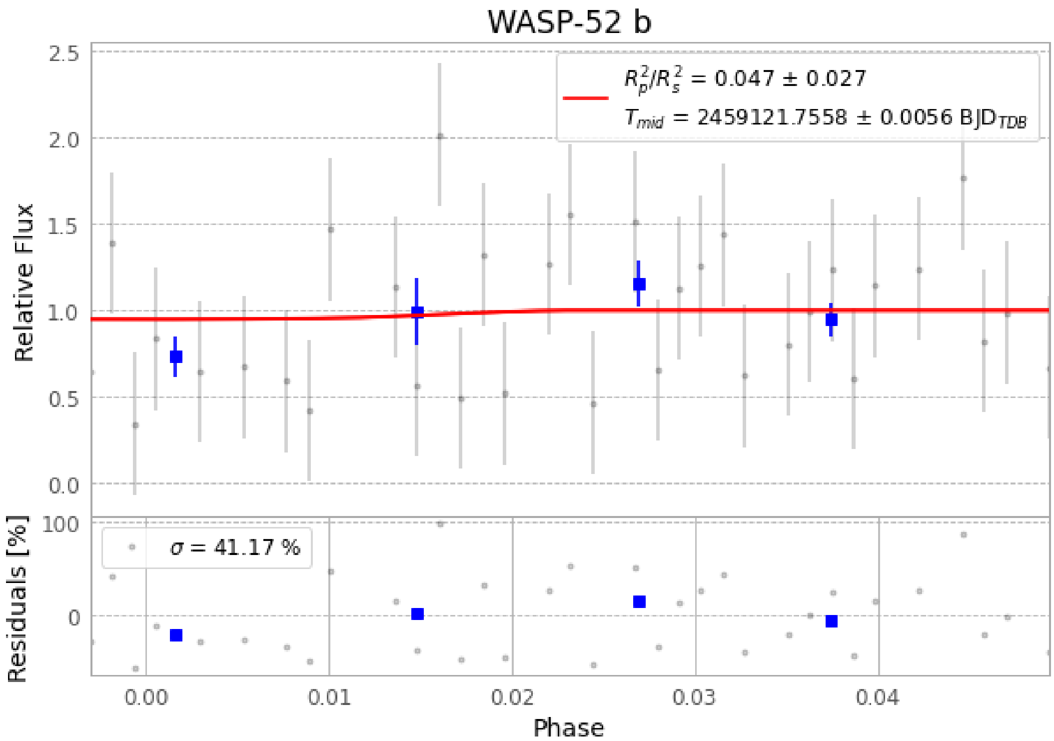

Figure 1 — FinalLightCurve_WASP-52 b_2020-09-28.png (SHA-256 6d65183f944b7e90…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 138], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f4ab669f4e6b639b…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

48 FITS frames (32 MB), `WASP-52200929055112.FITS` → `WASP-52200929081214.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-200929.log (cd524b3f7b04…), FinalLightCurve_WASP-52 b_2020-09-28.png (6d65183f944b…), FinalLightCurve_WASP-52 b_2020-09-28.csv (bfb7f6e41d5b…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)